DAY 33 · THEORY HANDOUT

Choose What Search Returns

Compare top-k, score filtering, MMR, and a saved index.

KGP Talkie

Scan to subscribe

Laxmi Kant Tiwari · kgptalkie.com

02 KGP Talkie

The idea and its useful terms

Compare top-k, score filtering, MMR, and a saved index.

Retriever

Returns documents for a question.

Top-k

Return up to k results ranked by the search.

MMR

Balance similarity with varied results.

The most similar two chunks may repeat the same line. MMR can include a different useful chunk so the context covers more of the question.

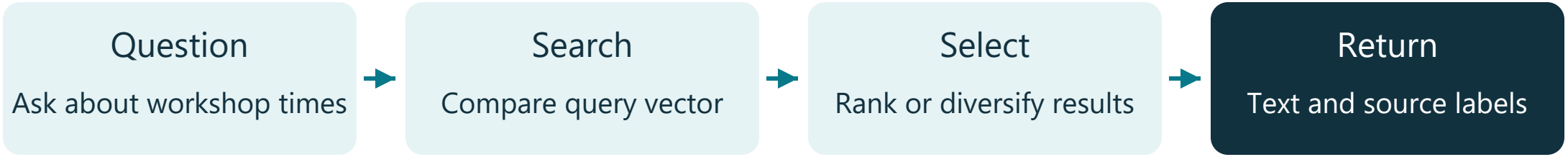
A score threshold can reject weak matches. Scores depend on the model and distance setting; do not treat a number as a universal quality mark.

> A saved FAISS bundle can contain a Python pickle. Load only files we created and trust.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

Similarity search ranks nearby vectors. MMR first fetches candidates and chooses a subset that balances closeness and variety. k is the final result count; fetch_k is the candidate count for MMR.

A saved index must match its embedding model and documents. Reusing it with changed inputs can give stale results. The distance score in our example is smaller for closer vectors; other scoring interfaces may differ.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

04 KGP Talkie

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Top-k

Take the nearest 2.

Can repeat one topic.

MMR

Choose useful variety.

Fetch candidates first.

Score filter

Drop weak matches.

Tune on sample questions.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> A saved FAISS bundle can contain a Python pickle. Load only files we created and trust.

05 KGP Talkie

Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

k = 2

Small reading context.

May miss another fact.

fetch_k = 4

MMR sees four candidates.

Chooses its final k.

Save and reopen

Avoid rebuilding each run.

Use only your own files.

Compare similarity and MMR on a set with repeated facts.

Print retrieved text and metadata before generating an answer.

Save an index you made, reload it, and repeat the query. Rebuild when the document or embedding model changes.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Save an index you made, reload it, and repeat the query. Rebuild when the document or embedding model changes.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> A saved FAISS bundle can contain a Python pickle. Load only files we created and trust.

DAY 33 · COMPLETE

Thank You

Follow KGP Talkie for new lessons, course updates and live sessions.

Scan to follow

Instagram
@kgptalkie

WhatsApp Channel
KGP Talkie